

MERN Full Stack Final Assessment (FA)

SET B — Issue / Bug Tracking System

API Response & Frontend Evaluator Contract

1. Purpose of This Document

This document defines the:

- expected API response formats
- status codes
- pagination structures
- authentication response structures
- analytics response formats
- frontend evaluator conventions
- Playwright compatibility

requirements for the:

Issue / Bug Tracking System — Set B

Students must strictly follow:

- route names
- response structures
- status codes
- data-testid conventions
- window.appState

structure mentioned in this document.

Failure to follow these conventions may affect automated evaluation.

2. Common Backend Response Standards

Success

Response

Status Code

200 / 201

Expected Structure

```
{
  "success": true,
  "message": "Operation successful",
  "data": {}
}
```



Error Response

Status Codes

Scenario	Status
Validation Error	400
Unauthorized	401
Forbidden	403
Not Found	404
Server Error	500

Expected Structure

```
{
  "success": false,
  "message": "Error message"
}
```

Pagination Response

Format Expected

```
{
  "success": true,
  "message": "Data fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

Health(users alone)

GET /health

```
{
  "success": true,
  "message": "Database connected successfully",
  "data": {
    "database": "connected",
    "documentCount": 62
  }
}
```

Authentication

POST /api/auth/register

POST /api/auth/login

GET /api/auth/me

Dataset Sync

POST /api/sync

```
{
  "success": true,
  "message": "Dataset synchronized successfully",
  "data": {
    "totalFetched": 421,
    "inserted": 334,
    "duplicates": 13,
    "rejected": 162
  }
}
```

Users

GET /api/users

```
{
  "success": true,
  "message": "Users fetched successfully",
  "total": 62,
  "data": [
    {
      "_id": "6a1f9f9148e26356bbe9b42c",
      "userId": "USR0001",
      "name": "System Admin",
      "email": "admin@test.com",
      "role": "admin",
      "department": "Administration",
      "status": "active",
      "createdAt": "2026-06-03T03:29:21.940Z",
      "updatedAt": "2026-06-03T03:29:21.940Z",
      "__v": 0
    },
  ]
}
```

GET /api/users/:userId

Example:

GET /api/users/USR1001

```
{
  "success": true,
  "message": "User fetched successfully",
  "data": {
    "_id": "6a1fa5649b17914ea7f1385a",
    "userId": "USR1001",
    "name": "Rahul Kumar",
    "email": "rahul.kumar@test.com",
    "role": "admin",
    "department": "Management",
    "status": "active",
    "createdAt": "2026-06-03T03:54:12.635Z",
    "updatedAt": "2026-06-03T03:54:12.635Z",
    "__v": 0
  }
}
```

Projects

POST /api/projects

```
{
  "success": true,
  "message": "Project created successfully",
  "data": {
    "_id": "6a1fa71d9b9485d6d635edb8",
    "projectId": "PROJ1",
    "title": "Issue Tracker Portal",
    "category": "Web Application",
    "description": "Issue tracking system",
    "owner": {
      "_id": "6a1fa5649b17914ea7f1385a",
      "userId": "USR1001",
      "name": "Rahul Kumar",
      "email": "rahul.kumar@test.com",
      "role": "admin"
    },
    "members": [
      {
        "_id": "6a1fa5649b17914ea7f1385b",
        "userId": "USR1002",
        "name": "Priya Iyer",
        "email": "priya.iyer@test.com",
        "role": "admin"
      },
      {
        "_id": "6a1fa5659b17914ea7f1385c",
        "userId": "USR1003",
        "name": "Arjun Nair",
        "email": "arjun.nair@test.com",
        "role": "admin"
      }
    ],
    "status": "active",
    "createdAt": "2026-06-03T04:01:33.821Z",
    "updatedAt": "2026-06-03T04:01:33.821Z",
    "__v": 0
  }
}
```

GET /api/projects

```
{
  "success": true,
  "message": "Projects fetched successfully",
  "total": 25,
  "data": [
    {
      "_id": "6a1fa5719b17914ea7f13897",
      "projectId": "PROJ1001",
      "title": "CRM Portal",
      "category": "CRM",
      "description": "Customer relationship management platform",
      "owner": {
        "_id": "6a1fa5659b17914ea7f1385d",
        "userId": "USR1004",
        "name": "Meera Krishnan",
        "email": "meera.krishnan@test.com",
        "role": "manager"
      }
    },
  ],
}
```

GET /api/projects/:projectId

```
{
  "success": true,
  "message": "Project fetched successfully",
  "data": {
    "_id": "6a1fa5719b17914ea7f13897",
    "projectId": "PROJ1001",
    "title": "CRM Portal",
    "category": "CRM",
    "description": "Customer relationship management platform",
    "owner": {
      "_id": "6a1fa5659b17914ea7f1385d",
      "userId": "USR1004",
      "name": "Meera Krishnan",
      "email": "meera.krishnan@test.com",
      "role": "manager"
    }
  },
}
```

PATCH /api/projects/:projectId



```
{
  "success": true,
  "message": "Project updated successfully",
  "data": {
    "_id": "6a1fa5719b17914ea7f13897",
    "projectId": "PROJ1001",
    "title": "CRM Portal",
    "category": "CRM",
    "description": "Customer relationship management platform",
    "owner": {
      "_id": "6a1fa5659b17914ea7f1385d",
      "userId": "USR1004",
      "name": "Meera Krishnan",
      "email": "meera.krishnan@test.com",
      "role": "manager"
    },
    "members": [
      {
        "_id": "6a1fa5669b17914ea7f13864",
        "userId": "USR1011",
        "name": "Ananya Raman",
        "email": "ananya.raman@test.com",
        "role": "developer"
      },
      {
        "_id": "6a1fa5669b17914ea7f13865",
        "userId": "USR1012",
        "name": "Surya Bose",
        "email": "surya.bose@test.com",
        "role": "developer"
      },
      {
        "_id": "6a1fa5679b17914ea7f13866",
        "userId": "USR1013",
        "name": "Keerthi Sharma",
        "email": "keerthi.sharma@test.com",
        "role": "developer"
      },
      {
        "_id": "6a1fa5679b17914ea7f13867",
        "userId": "USR1014",
        "name": "Rohit Kumar",
        "email": "rohit.kumar@test.com",
        "role": "developer"
      }
    ]
  }
}
```



```
    },
    {
      "_id": "6a1fa56d9b17914ea7f13887",
      "userId": "USR1046",
      "name": "Aditi Krishnan",
      "email": "aditi.krishnan@test.com",
      "role": "tester"
    },
    {
      "_id": "6a1fa56d9b17914ea7f13888",
      "userId": "USR1047",
      "name": "Varun Menon",
      "email": "varun.menon@test.com",
      "role": "tester"
    }
  ],
  "status": "active",
  "startDate": "2026-01-10T00:00:00.000Z",
  "createdAt": "2026-06-03T03:54:25.176Z",
  "updatedAt": "2026-06-03T03:54:25.176Z",
  "__v": 0
}
```

DELETE /api/projects/:projectId

Examples:

GET /api/projects/PROJ1001

PATCH /api/projects/PROJ1001

DELETE /api/projects/PROJ1001

Issues

POST /api/issues

```
{
  "success": true,
  "message": "Issue created successfully",
  "data": {
```



```

    "_id": "6a1fab9608c245f885c7e474",
    "issueId": "ISS1",
    "title": "Login Page Crash",
    "description": "Application crashes after login",
    "project": {
      "_id": "6a1fa5719b17914ea7f13897",
      "projectId": "PROJ1001",
      "title": "CRM Portal"
    },
    "assignedTo": {
      "_id": "6a1fa5669b17914ea7f13865",
      "userId": "USR1012",
      "name": "Surya Bose",
      "email": "surya.bose@test.com",
      "role": "developer"
    },
    "reportedBy": {
      "_id": "6a1f9f9148e26356bbe9b42c",
      "userId": "USR0001",
      "name": "System Admin",
      "email": "admin@test.com"
    },
    "priority": "high",
    "severity": "critical",
    "status": "open",
    "createdAt": "2026-06-03T04:20:38.128Z",
    "updatedAt": "2026-06-03T04:20:38.128Z",
    "__v": 0
  }
}

```

GET /api/issues

```

{
  "success": true,
  "message": "Issues fetched successfully",
  "total": 182,
  "data": [
    {
      "_id": "6a1fa5789b17914ea7f138af",
      "issueId": "ISS1001",
      "title": "Lead assignment failure",
      "description": "",
      "project": {

```

```

        "_id": "6a1fa5719b17914ea7f13897",
        "projectId": "PROJ1001",
        "title": "CRM Portal",
        "status": "active"
    },
    "assignedTo": {
        "_id": "6a1fa5669b17914ea7f13864",
        "userId": "USR1011",
        "name": "Ananya Raman",
        "email": "ananya.raman@test.com",
        "role": "developer"
    },
    "reportedBy": {
        "_id": "6a1fa56d9b17914ea7f13887",
        "userId": "USR1046",
        "name": "Aditi Krishnan",
        "email": "aditi.krishnan@test.com",
        "role": "tester"
    },
    "priority": "high",
    "severity": "major",
    "status": "open",
    "createdAt": "2026-06-03T03:54:32.636Z",
    "updatedAt": "2026-06-03T03:54:32.636Z",
    "__v": 0
  },
}
}

```

GET /api/issues/:issueId

```

{
  "success": true,
  "message": "Issue fetched successfully",
  "data": {
    "_id": "6a1fa5789b17914ea7f138af",
    "issueId": "ISS1001",
    "title": "Lead assignment failure",
    "description": "",
    "project": {
      "_id": "6a1fa5719b17914ea7f13897",
      "projectId": "PROJ1001",
      "title": "CRM Portal",
      "status": "active"
    },
    "assignedTo": {

```

```

      "_id": "6a1fa5669b17914ea7f13864",
      "userId": "USR1011",
      "name": "Ananya Raman",
      "email": "ananya.raman@test.com",
      "role": "developer"
    },
    "reportedBy": {
      "_id": "6a1fa56d9b17914ea7f13887",
      "userId": "USR1046",
      "name": "Aditi Krishnan",
      "email": "aditi.krishnan@test.com",
      "role": "tester"
    },
    "priority": "high",
    "severity": "major",
    "status": "open",
    "createdAt": "2026-06-03T03:54:32.636Z",
    "updatedAt": "2026-06-03T03:54:32.636Z",
    "__v": 0
  }
}

```

PATCH /api/issues/:issueId

```

{
  "success": true,
  "message": "Issue updated successfully",
  "data": {
    "_id": "6a1fa5789b17914ea7f138af",
    "issueId": "ISS1001",
    "title": "Login Failure Bug",
    "description": "",
    "project": {
      "_id": "6a1fa5719b17914ea7f13897",
      "projectId": "PROJ1001",
      "title": "CRM Portal"
    },
    "assignedTo": {
      "_id": "6a1fa5669b17914ea7f13864",
      "userId": "USR1011",
      "name": "Ananya Raman",
      "email": "ananya.raman@test.com",
      "role": "developer"
    }
  }
}

```

```
{
  "reportedBy": {
    "_id": "6a1fa56d9b17914ea7f13887",
    "userId": "USR1046",
    "name": "Aditi Krishnan",
    "email": "aditi.krishnan@test.com",
    "role": "tester"
  },
  "priority": "high",
  "severity": "major",
  "status": "open",
  "createdAt": "2026-06-03T03:54:32.636Z",
  "updatedAt": "2026-06-03T04:21:33.704Z",
  "__v": 0
}
```

DELETE /api/issues/:issueId

Examples:

GET /api/issues/ISS1001

PATCH /api/issues/ISS1001

DELETE /api/issues/ISS1001

Issue Workflow

Assign Issue

PATCH /api/issues/:issueId/assign

Example:

PATCH /api/issues/ISS1001/assign

```
{
  "success": true,
  "message": "Issue assigned successfully",
  "data": {
    "_id": "6a1fa5789b17914ea7f138af",
    "issueId": "ISS1001",
    "title": "Login Failure Bug",
    "description": "",
    "project": {
```

```
      "_id": "6a1fa5719b17914ea7f13897",
      "projectId": "PROJ1001",
      "title": "CRM Portal"
    },
    "assignedTo": {
      "_id": "6a1fa5669b17914ea7f13865",
      "userId": "USR1012",
      "name": "Surya Bose",
      "email": "surya.bose@test.com",
      "role": "developer"
    },
    "reportedBy": {
      "_id": "6a1fa56d9b17914ea7f13887",
      "userId": "USR1046",
      "name": "Aditi Krishnan",
      "email": "aditi.krishnan@test.com",
      "role": "tester"
    },
    "priority": "high",
    "severity": "major",
    "status": "open",
    "createdAt": "2026-06-03T03:54:32.636Z",
    "updatedAt": "2026-06-03T04:21:59.930Z",
    "__v": 0
  }
}
```

Change Status

PATCH /api/issues/:issueId/status

Example:

PATCH /api/issues/ISS1001/status

```
{
  "success": true,
  "message": "Issue status updated successfully",
  "data": {
    "issueId": "ISS1001",
    "previousStatus": "open",
    "newStatus": "in-progress"
  }
}
```

Comments

POST /api/comments

```
{
  "success": true,
  "message": "Comment added successfully",
  "data": {
    "commentId": "COM1",
    "issue": "6a1fa5789b17914ea7f138af",
    "user": "6a1f9f9148e26356bbe9b42c",
    "message": "Issue verified successfully",
    "_id": "6a1fac0008c245f885c7e477",
    "createdAt": "2026-06-03T04:22:24.075Z",
    "updatedAt": "2026-06-03T04:22:24.075Z",
    "__v": 0
  }
}
```

GET /api/comments

```
{
  "success": true,
  "message": "Comments fetched successfully",
  "total": 25,
  "data": [
    {
      "_id": "6a1fa5c99b17914ea7f13964",
      "commentId": "COM1001",
      "issue": {
        "_id": "6a1fa5789b17914ea7f138af",
        "issueId": "ISS1001",
        "title": "Login Failure Bug"
      },
      "user": {
        "_id": "6a1fa56d9b17914ea7f13887",
        "userId": "USR1046",
        "name": "Aditi Krishnan",
        "email": "aditi.krishnan@test.com"
      },
      "message": "Issue reproduced successfully",
    }
  ]
}
```

```
      "createdAt": "2026-06-01T09:15:00.000Z",
      "updatedAt": "2026-06-01T09:15:00.000Z",
      "__v": 0
    },
  ]
}
```

GET /api/comments/:commentId

```
{
  "success": true,
  "message": "Comment fetched successfully",
  "data": {
    "_id": "6a1fa5c99b17914ea7f13964",
    "commentId": "COM1001",
    "issue": {
      "_id": "6a1fa5789b17914ea7f138af",
      "issueId": "ISS1001",
      "title": "Login Failure Bug"
    },
    "user": {
      "_id": "6a1fa56d9b17914ea7f13887",
      "userId": "USR1046",
      "name": "Aditi Krishnan",
      "email": "aditi.krishnan@test.com"
    },
    "message": "Issue reproduced successfully",
    "createdAt": "2026-06-01T09:15:00.000Z",
    "updatedAt": "2026-06-01T09:15:00.000Z",
    "__v": 0
  }
}
```

DELETE /api/comments/:commentId

Examples:

GET /api/comments/COM1001

DELETE /api/comments/COM1001

Analytics

GET /api/analytics/issues

```
{
```

```
{
  "success": true,
  "message": "Issue analytics fetched successfully",
  "data": {
    "totalIssues": 182,
    "openIssues": 49,
    "inProgressIssues": 28,
    "testingIssues": 37,
    "resolvedIssues": 44,
    "closedIssues": 24
  }
}
```

GET /api/analytics/projects

```
{
  "success": true,
  "message": "Project analytics fetched successfully",
  "data": {
    "activeProjects": 17,
    "completedProjects": 6,
    "archivedProjects": 2,
    "projects": [
      {
        "projectId": "PROJ1001",
        "title": "CRM Portal",
        "owner": "Meera Krishnan",
        "status": "active",
        "issueCount": 10
      },
      {
        "projectId": "PROJ1002",
        "title": "Lead Management System",
        "owner": "Meera Krishnan",
        "status": "active",
        "issueCount": 7
      }
    ]
  }
}
```

GET /api/analytics/developers

```
{
  "success": true,
  "message": "Developer analytics fetched successfully",
  "data": {
    "highestResolvedDeveloper": {
```



```
    "userId": "USR1024",
    "name": "Sanjana Reddy",
    "resolvedIssues": 6
  },
  "developers": [
    {
      "userId": "USR1011",
      "name": "Ananya Raman",
      "assignedIssues": 2,
      "resolvedIssues": 0
    },
    {
      "userId": "USR1012",
      "name": "Surya Bose",
      "assignedIssues": 4,
      "resolvedIssues": 0
    }
  ],
  ]}
```

Filtering, Search & Pagination

Issues

GET /api/issues?status=open

GET /api/issues?priority=high

GET /api/issues?severity=critical

GET /api/issues?search=login

GET /api/issues?page=1&limit=10

GET /api/issues?status=open&priority=high&page=1&limit=10

Projects

GET /api/projects?status=active

GET /api/projects?category=Web Application



GET /api/projects?owner=Rahul

GET /api/projects?search=tracker

GET /api/projects?page=1&limit=10

Comments

GET /api/comments?issueId=ISS1001

GET /api/comments?search=login

GET /api/comments?page=1&limit=10

Total Endpoint Count

Health	1
Authentication	3
Sync	1
Users	2
Projects	5
Issues	5
Issue Workflow	2
Comments	4
Analytics	3

Total	26 Endpoints



3.